

Enhancing climate resilience in businesses: The role of artificial intelligence

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Abstract

The abrupt rise in extreme weather events (floods, heat waves, droughts, etc.) due to changing climate in the last decades has increased the level of threats to various sectors (agriculture, energy, transportation, etc.) globally. The climate projections from global circulation models indicate even more intense and frequent extreme events in the future, which in turn pose more risks to socioeconomic infrastructure. The enhanced understanding of the climate-related financial risk associated with businesses has driven efforts to include critical information on probable risks associated with climate change in financial decision-making. In this study, we have presented a framework to assess the need of incorporating climate risk assessment as an integral part of business operations. We also reviewed revealed literature to understand the possible impacts of climate change on various sectors and presented key strategies to assess the climate risk associated with them. Also, a framework incorporating probable climate threats to business ecology with principles of robustness, resourcefulness, redundancy, and rapidity has been proposed to adapt and mitigate associated risks for a climate-resilient business ecosystem. The integration of Artificial Intelligence in managing risk could be a promising tool for enhancing business resilience to climate change and could be used as a tool. Robust and accurate predictions of climate and weather extremes from deep learning algorithms at a significant lead time can help in minimizing the associated risk with a business infrastructure. Atmospheric Rivers (ARs), a weather extreme cause huge socioeconomic risk by triggering floods and droughts in various continents of mid-latitude regions. We have presented a case study investigating the ability of deep learning algorithms to predict ARs. The results from the analysis advocate the application of deep learning algorithms to predict weather and climate extremes in decision support systems to enhance the climate resilience of a business ecosystem.



Introduction

Climate change is an indisputable worldwide concern with far-reaching effects for both natural and human systems. The impacts have been more noticeable over the past ten years, evidenced by the increase in the frequency and severity of extreme weather events (Adger et al., 2005; Biesbroek et al., 2013; Fiedler et al., 2021; Izaguirre et al., 2021). Hurricanes, floods, wildfires, and droughts are only a few examples of events that have severely damaged vital infrastructure, disrupted economies, and resulted in huge costs to society and the environment (Kumar et al., 2021). Given these mounting threats, it is critical for businesses to understand the effects of climate change on their operations and build resilience plans. The business sector plays a crucial role in society and is intricately linked to various aspects of the economy, environment, and social fabric. Climate change poses significant risks to businesses, ranging from supply chain disruptions and increased operational costs to reduced demand for goods and services (Campiglio, 2018; Lamperti et al., 2019). It is estimated that for each 1 °C increase in average temperature, the global GDP decreases by around 1.2% emphasizing the urgent need for effective climate change mitigation and adaptation measures to mitigate the adverse effects of rising temperatures on global economic productivity (Burke et al., 2015). Schlenker and Roberts (2009) highlight the severe damages that climate change can inflict on U.S. crop production and calls for proactive measures to safeguard food security with changing climatic conditions. These risks necessitate proactive measures to adapt and mitigate the effects of climate change, ensuring business continuity and long-term sustainability.

To effectively address climate risks, businesses need to integrate climate science into their conventional decision-making processes. This integration enables them to assess their sensitivity to climate change, anticipate future impacts, and plan to improve resilience. It is, however, difficult to incorporate climate science into business operations. Direct use of climate data by businesses is challenging due to the complexity of climate models, their spatial and temporal resolutions, and the uncertainty surrounding future estimates (Center for Climate and Energy Solutions, 2008). Therefore, it is required to introduce novel approaches that can bridge the gap between climate science and business decision-making.

Artificial intelligence (AI) is the emulation of human intellect in machines, allowing them to learn from data, make informed decisions, and perform tasks autonomously (Alzubaidi et al., 2021). Deep learning, a branch of AI, employs artificial neural networks with numerous layers to analyze complex patterns and correlations in data (Hao et al., 2016; Lecun et al., 2015). It has demonstrated impressive performance in a number of fields, including image identification, natural language processing, and now climate research. The integration of AI and deep learning techniques in managing risk could be a promising tool for enhancing business resilience to climate change (Papadopoulos and Balta, 2022; Rocha Rodrigues et al., 2018). Taking advantage of big data analytics and machine learning algorithms, businesses can extract important insights from vast climate datasets, enabling them to assess vulnerability and identify potential risks and impacts (Perera et al., 2020). AI-driven risk modeling and predictive analytics help organizations anticipate and manage climate-related risks such as supply chain interruptions, market upheavals, and regulatory changes. Furthermore, given climate uncertainty and variability, AI-powered decision support systems can optimize resource allocation and operational efficiency. Advanced forecasting models driven by AI and deep learning are a remarkable achievement in enhancing climate resilience. Deep learning algorithms can analyze climate data and patterns to generate accurate predictions for future climate phenomena, such as floods, droughts, cyclones, and other extreme weather events (Scher and Messori, 2018). In a similar way, deep learning algorithms can also be employed to predict the availability of Atmospheric Rivers (ARs), which have been associated with heavy

rainfall and flood events in many mid-latitude regions of the world. Accurate forecasts of ARs can significantly aid in flood and water resource management, allowing businesses and communities to take proactive measures and reduce potential damages.

While the integration of AI and deep learning presents exciting opportunities for enhancing business resilience, it also brings some challenges along with. AI-driven approaches essentially require high-quality data, robust algorithms, and skilled personnel for successful implementation (Araos et al., 2017; Ford, 2018). Collaborations between climate scientists, AI experts, and business stakeholders are crucial to developing tailored solutions and effectively utilizing AI outputs for decision-making. As climate change continues to pose substantial threats to businesses, there is a rising demand for novel approaches that combine climate research and AI-powered solutions. Artificial intelligence (AI) and deep learning provide useful tools for identifying vulnerabilities, reducing risks, and strengthening the resilience of a business (Ben Ayed and Hanana, 2021; Papadopoulos and Balta, 2022). Taking advantage of recent climate data, big data analytics, and advanced machine learning algorithms, businesses can make informed decisions, optimize operations, and adapt to changing climate conditions. Embracing AI-driven recommendations will not only assist businesses in navigating climate change threats but will also contribute to a more sustainable and resilient future.

This study provides a novel framework to understand the vulnerability and risk associated with businesses due to climate change. The prime focus of the study is to understand the possible risks to businesses and the significance of taking action to mitigate those risks, which is a crucial step in addressing these climate consequences. This paper also presents a framework to analyze the degree of vulnerability of a business and recognize the need of taking necessary measures for climate change adaptation and mitigation. We also discussed strategies to be incorporated and presented a framework to make a climate-resilient business ecosystem. Also, a Deep Learning approach has been presented as a case study to forecast ARs, which could be integrated into a decision support system along with other relevant hazards to reduce the risk posed to the business environment. The study will provide valuable insights to bring resilience in businesses after understanding the long-term and short-term impacts of climate change and assessing the vulnerability (risk).

Section snippets



Literature review

Climate change poses both risks and opportunities for businesses across various sectors (Papadopoulos and Balta, 2022). By studying climate change, businesses can understand the potential impacts on their operations, supply chains, and markets which may lead to supply chain disruptions, increased costs, damage to infrastructure, and reputational harm (Manzione Filho, 2022). Understanding these risks helps businesses develop effective adaptation and resilience strategies. Moreover, studying ...

Methodology

The manuscript is composed of two major parts: (a) Reviewing research articles and reports to develop a framework to analyze the climate risk and enhance resilience for a business ecosystem and (b) a case study focusing on the scope of artificial intelligence to minimize the risk associated with climate extremes.

In the first part of this paper, a methodology was employed to develop a comprehensive framework for assessing the climate risk of businesses and enhancing their resilience. The ...

Climate resilient business ecosystem

The ability of a system to lessen the likelihood of a disruption, to absorb it if it occurs, and to recover rapidly just after that is known as resilience (Argyroudis et al., 2022; Folke, 2002). A system can be robust if it is capable of reducing the likelihood of catastrophe (reduction), reducing the repercussions of a failure (absorption), and reducing the time required to restore to its normal state (recovery). The resilience of an ecosystem can be measured through robustness, the power or ...

A case study: application of artificial intelligence in early warning systems

Atmospheric rivers (ARs) are filamentary regions of high-water vapor flux in the lower troposphere (Ionita et al., 2020; Ralph and Dettinger, 2011). These ARs, usually associated with low-level jets, exist within the warm conveyor belts (WCBs) of the mid-latitude region and play a vital role in transporting moisture to the poles (Paltan et al., 2017; Ralph et al., 2006). It has been estimated that approximately 90% of the poleward moisture transport is being transported by these ARs in the ...

Conclusion

Climate change poses significant risks to businesses beyond the direct effects of weather on infrastructure and operations. While some leading businesses are starting to evaluate other climate-related risks, such as shareholder and consumer views, compliance with climate legislation, and the impact of greenhouse gas limits, many businesses still overlook the physical risks associated with climate change. This is partly due to the long-term nature of climate shifts and the uncertainty ...

Limitations and future scope

While the framework presented in this study provides a thorough approach to analyzing climate risk and improving business resilience, it is also important to understand its limitations. The framework presented is theoretical and relies on the availability and quality of data (information). The accuracy and reliability of the assessment may get affected by the data gaps and limitations in certain regions or sectors. To better comprehend the past and future effects of climate change, efforts ...

CRedit authorship contribution statement

Shivam Singh: Conceptualization, Methodology, Data curation, Software, Writing – original draft, and Revision, Visualization, Investigation. **Manish Kumar Goyal:** Conceptualization, Writing – review & editing, and Revision, Validation, Visualization, Investigation. ...

Declaration of competing interest

We hereby declare that the authors do not possess any competing interests related to this manuscript entitled, "**Enhancing Climate Resilience in Businesses: The Role of Artificial Intelligence**" submitted in the Journal of 'Journal of Cleaner Production'. ...

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